Project status report

This report sums up the work of the last quarter for each team. Every team writes a short card; the cards sit in a two-column grid, and a row of cards that does not fit in the room left on a page breaks at the page end, the way a browser prints it.

Rows are only moved to the next page whole when one of their cards asks for it, with break-inside: avoid, or when a card's paragraph cannot leave enough lines behind for its orphans setting.

Teams

Layout

The layout team finished the work on line breaking next to inline blocks, images and form controls, and reworked how fieldset legends are placed on the border.

Fonts

Variable fonts are now instanced before subsetting, and the bundled fonts gained the missing currency signs.

Rendering

The renderer draws dashed strokes and text on a path. Shading patterns are faster, and the anti-aliasing of thin lines matches other viewers more closely.

Next quarter the team looks at blend modes in knockout groups and at soft masks with transfer functions.

Signing

Long-term validation data is now embedded for every certificate in the chain, and timestamps from a second authority can be added later.

Accessibility

Tagged output covers tables with row headers, lists nested in tables and figures with captions. The PDF/UA check runs on every sample document in CI.

Remaining work: alternative text for SVG and the reading order of floated boxes.

The team also reviewed the structure of form fields and how their labels are announced by screen readers.

Markdown

Tables with alignment, footnotes and task lists follow the CommonMark and GFM rules.

Performance

Conversions of long tables are a fifth faster, mostly from caching text

Hostile input

The fuzzer ran for two weeks without new findings. Limits on nesting depth, image

measurements per font and size and from reusing line boxes between passes.

Memory use stays flat with the number of pages.

size and font tables are counted before allocating.

Milestones

The milestones of the quarter, one card a month. Each release went through two release candidates, and every candidate was tested against the golden documents and the conformance samples before it was published. Two of the candidates were held back for a week, both for fixes to the line breaking of right-to-left text. The release card asks to be kept in one piece, and its row is not broken at the end of a page but moved to the next one.

July

Version 1.9 shipped with the new hyphenation dictionaries and the first part of the work on grid layout across pages.

August

Version 1.10 added signing with long-term validation and the PDF/A-3 output for electronic invoices.

September

Version 1.11 brought variable fonts, text on a path in SVG and many fixes to the line breaking of mixed scripts.

October

The work on fieldsets, inline borders and grid rows was merged.

November (planned)

Version 1.12 with the October layout work, and a first look at running elements in page margins.

December (planned)

A quiet month: fixes only, and the yearly review of the third-party notices and bundled data.

January (planned)

Work starts on tables that repeat their footers on every page.

February (planned)

Version 1.13, if the table work is ready in time.

March (planned)

Conformance checks for PDF/A-4.

April (planned)

Planning for the next year.

Release (kept whole)

This card has break-inside: avoid, so its row moves to the next page whole instead of breaking at the page end.

Release candidates go out every second week, and the final release follows when no regressions are open.

The changelog lists every visible change.

Documentation

The manual gained pages on page breaks, running headers and form fields.

Risks

Fonts on servers

Servers often have few fonts installed. The bundled fonts cover Latin, Greek and Cyrillic; other scripts need fonts supplied by the caller.

A missing glyph is taken from the fallback list, and a warning names the character.

Browser differences

Chrome is the reference for layout. Differences found in reports are fixed one at a time, with a test page for each.

Key figures

Releases	Three minor releases and five patch releases.
Issues	142 opened and 151 closed, of which 38 were layout differences from browsers.
Contributors	Eleven people sent at least one change.
Test documents	Twelve new golden documents and four new hostile inputs.
Support	212 questions answered, most within a day.
Fuzzing	Two weeks of mutation fuzzing on every parser without a new finding.

Support

The support card asks for at least four lines of its first paragraph on a page (orphans: 4). When fewer fit at the end of a page, its row moves to the next page whole, although the card next to it would fit. Without that setting the row would break between two lines of the paragraph, as the rows of the team grid do. The card on the right is short and would fit on this page by itself, but a row is kept together: its cards start on the same page.

Support (orphans: 4). Questions from users are answered within two working days. Most of them are about fonts that are not installed on the server, page sizes and how to load images from a web address through a resource resolver.

Frequent answers become pages in the manual.

Website

The online converter shows the settings as code that can be copied into a project.

Outlook

The coming quarter is about tables and page margins. Long tables already repeat their headers on every page; they will also repeat their footers, and a row that is too tall for a page will break between its lines instead of being cut. Running elements in the page margins will let a document put its own title and chapter name at the top of every page without a separate header template.

The second theme is speed. Most documents convert in a few milliseconds, but documents with thousands of table rows spend most of their time measuring the same text again. A cache of measured runs per font and size should halve that time without changing a single byte of the output.

The third theme is the comparison with browsers. Every difference reported by users is reduced to a small test page, compared with Chrome word by word, and kept as a golden document once it is fixed, so it cannot come back unnoticed. This report is one of those documents: its grids break across pages the way Chrome prints them.

None of this changes the public interface. New settings get new properties with defaults that keep today's output, and every change to the output of an existing document is listed in the release notes together with the golden documents it changed and the reason. Callers that cache converted documents and compare them byte for byte can rely on that: a document converted with the same version and the same settings comes out the same, on every machine and in every culture, and a new version changes it only where the release notes say so.

Finally, the documentation will gain a page on page breaks that explains break-before, break-after and break-inside, orphans and widows, and how rows of tables and grids are split.

Next steps

Tables

Repeat table footers on every page, break tall rows between their lines, and keep a caption with the first rows of its table.

Page margins

Running elements for titles and chapter names, page counters in any margin box, and leaders in a table of contents.

Owner: the layout team, with help from the documentation team for the

Owner: the layout team. Review in the second week of the quarter, release with version 1.13.

examples.

Speed

A cache of measured text runs per font and size.

Documentation

A page on page breaks, orphans and widows.